Spontaneous Capillary-Inertial Dewetting at the Microscopic Scale

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We resolve the dewetting dynamics of water films on partially wetting surfaces at the microscopic scale and highlight its distinctions from liquid wetting. Fast dewetting occurring at the millisecond scale is dominated by capillary and inertial forces, obeying a power law with a wettability-independent exponent, which is different from the capillary-inertial dynamic wetting. Molecular-level inspections of moving contact lines further reveal that liquid molecules retract from solid surfaces through a rolling motion, whereas dynamic wetting proceeds mainly with the sliding motion. These distinct hydrodynamic behaviors and molecular kinetics explicitly suggest that dynamic dewetting is not simply the reverse of dynamic wetting, which has significant implications for interfacial fluid physics.

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Capillary wetting and dewetting play essential roles in small-scale fluidic devices and other interface-dominated systems [1,2] and, generally, appear in succession. In a wetting process, a fluid advances on a solid surface to maximize contact [3,4], which is beneficial for enhancing thermal cooling at high temperatures [5,6] and controlling leaf retention of pesticide solutions [7]. A dewetting process, by contrast, proceeds by retracting the liquid from the solid—a phenomenon that seems to be the reversal of liquid wetting and has profound implications for self-cleaning surfaces [8,9], anti-icing coatings [10], and, recently, hydrovoltaic electricity generators [9,11,12]. Motivated by both fundamental importance and significant application interest, understanding the dynamics of how liquids wet and dewet solid surfaces has been a long-standing quest in the field of fluid physics [2,13–16].

Early studies on spontaneous, capillary-driven wetting of low-viscosity liquids date back to the 1910s [17], but experimentally resolving the dynamics in the full time domain has only become accessible over the past few decades [2,3]. Upon contact, by either dipping a solid plate into a liquid [18,19] or depositing a liquid droplet onto a solid surface [20,21], the liquid rapidly spreads out driven by interfacial forces [22,23]. At the beginning, the wetting process is mainly resisted by the inertia of the liquid, and the balance between capillary and inertial forces or energies yields wettability-dependent scaling laws in time for moving contact lines, which persist for a few milliseconds in millimeter-sized systems (i.e., a few times of the characteristic capillary-inertial time τ_i [20,21,24,25]). Note that at very early times after contact (typically less than $0.1\tau_\mathrm{i}$ on hydrophobic surfaces) liquid wetting would obey a wettability-independent power law [22,23]. With the ongoing spreading, the contact line velocity slows down, and accordingly, the inertial effect lessens, giving way to other forces governing the dynamics. Viscous friction in the liquid rim [2,26,27] and molecular dissipation in the three-phase confluence region [28,29] were widely recognized as two main dissipating channels during the subsequent wetting, which can last up to hours [3]. Recent molecular dynamics (MD) simulations further revealed that liquid molecules near the contact line proceed in a rolling manner immediately after liquid–solid contact and shortly afterward transit to the sliding mode [14]. In striking contrast to well-understood dynamic wetting, the dewetting of liquids from solids has received much less attention [2,13]. This disparity may be attributed to the intuitive expectation of a strong similarity in the dynamics between wetting and dewetting, both of which seem to be reverse counterparts [3,30,31]. Although some hydrodynamic models of liquid wetting have been adapted to describe dewetting phenomena such as film rupture [30,32] and contraction [31] on partially wetting surfaces, importantly, physical processes such as liquid wetting and dewetting always involve energy dissipation (though it might be very low) and are inherently irreversible from the thermodynamic perspective. Therefore, dewetting should not

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presumably be considered as the reverse of a wetting process in any given liquid–solid systems. More investigations are thus required to understand the detailed dynamics of liquid dewetting and how it differs from liquid wetting.

In this Letter, we report explicit evidence that dynamic dewetting is physically not the reverse of dynamic wetting by investigating the spontaneous dewetting of thin water films from partially wetting solid surfaces at the microscopic scale—a liquid–solid system that can be frequently encountered in nature [33,34] and microengineering systems [35,36]. Via experiments, we demonstrate that fast dewetting, which occurs on the millisecond timescale, is dominated by capillary and inertial forces; however, it obeys a wettability-independent scaling law differing from that of dynamic wetting [20,21]. Using MD simulations, we reproduce the main dynamics of the capillary-inertial dewetting, which allows us to further probe the liquid motion at molecular resolution. We found that liquid retracts solely through a rollinglike motion, which is distinct from the sliding-dominated advancing liquid [14].

Dewetting experiments were performed by evaporating millimeter-sized water droplets on hydrophilic glass surfaces (of equilibrium water contact angle $\theta_{\mathrm{eq}}^{\mathrm{Phi}} = 42 \pm 2°$) decorated with regular arrays of relatively hydrophobic circular zones ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 80 \pm 1°$ and $102 \pm 1°$) under standard laboratory conditions [Fig. 1(a)]. These hybrid hydrophilic–hydrophobic surfaces were fabricated through a photolithography process, followed by selective surface hydrophobization. The radii R_0 of hydrophobic zones varied from 10 to 200 μm, while their spacing (S) was adjusted between 160 and 3200 μm, corresponding to hydrophobic-to-hydrophilic area ratios $\varphi \in [0.01, 0.07]$. In each test, a millimeter-sized water droplet was dripped onto the solid surface, which first spread out for a few seconds and then shrank after evaporating for 10–20 min. The motion of the receding liquid was monitored using a high-speed camera with a recording rate of 160,000 fps under an inverted microscope [Fig. 1(a)]. For experimental details, see Supplemental Material (SM) [37], Sec. I.

Because of the low area ratio φ, patterning hydrophilic surfaces with hydrophobic zones does not notably change the wetting properties of the original surfaces, as theoretically predicted and experimentally confirmed (Table S1 [37]). As such, an evaporating water droplet maintained its contact line pinned on the hydrophilic region of the hybrid surfaces until the liquid dried into a thin film with a contact angle of ~8° [Fig. 1(b)]. The subsequent motion of the receding contact line on the hydrophilic region was slow, with a maximum velocity in the micrometers/second range. Such slow dewetting is predominantly driven by liquid evaporation rather than hydrodynamic forces [43].

When the contact line reached the circular hydrophobic zone, fast dewetting was triggered. As an illustrative example, in Fig. 1(c), we show selected snapshots of a receding contact line moving across a hydrophobic zone

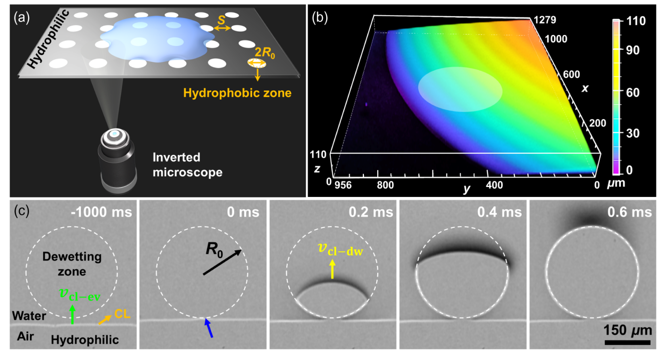

FIG. 1. (a) Sketch of the experimental setup. (b) Three-dimensional confocal microscopy image of a water film before dewetting from a hydrophobic zone with $R_0 = 200$ μm and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$. The white circle represents the hydrophobic zone. (c) Snapshots of a contact line moving across a circular hydrophobic zone with $R_0 = 150$ μm and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$.

with $R_0 = 150$ μm and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$. Initially, the contact line was straight and moved uniformly on the hydrophilic region with an average velocity $\bar{v}_{\mathrm{cl-ev}} \approx 16$ μm/s. Once the contact line passed through the hydrophilic–hydrophobic boundary, a distinctive dewetting scenario unfolded. Starting from the intersection point of the straight contact line with the hydrophobic circular zone [cf. the blue arrow in Fig. 1(c) and Movie 1], water radially retracted upward into the thin film under the capillary force, forming a growing dry patch on the solid surface. Liquid dewetting was completely constrained in the defined hydrophobic zone, and the receding contact line always took the shape of a circular arc (SM [37], Sec. II), that is, the contact line motion proceeded in a self-similar manner. Moreover, water withdrawing from the dried patch collected at the receding rim, and its high-curvature surface (defined by the receding contact angle $\theta_{\mathrm{r}}^{\mathrm{Pho}} \approx 87°$) caused the appearance of a dark shadow around the moving contact line in the acquired images [31,44]. The entire process lasted $t_{\mathrm{dw}} \approx 0.59$ ms and was independent of $\bar{v}_{\mathrm{cl-ev}}$ in the experimental range of 0.3–19 μm/s (SM [37], Sec. II). The average dewetting velocity was estimated to be $v_{\mathrm{cl-dw}} \approx 2R_0/t_{\mathrm{dw}} \approx 0.5$ m/s, which is six orders of magnitude faster than $\bar{v}_{\mathrm{cl-ev}}$. Such dewetting behavior was also observed on other hydrophilic surfaces patterned with circular hydrophobic zones of different sizes (SM [37], Sec. III), and it is independent on the geometric shape of hydrophobic zones (Appendix A).

To quantitatively evaluate the fast dewetting process, we analyzed the temporal evolution of the growing dry area A, and the results acquired for 12 circular "hydrophobic" zones with two different wetting properties ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and $102°$) and six radii ($R_0 = 10$–200 μm) are comparatively displayed in Fig. 2(a). A nonlinearly increases with time t, and the dewetting process is apparently shorter for water films on hydrophobic zones with a smaller area ($A_0 = \pi R_0^2$) or a larger $\theta_{\mathrm{eq}}^{\mathrm{Pho}}$. Previous

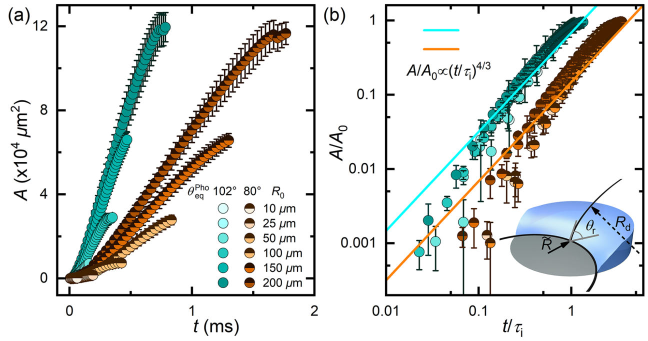

FIG. 2. (a) The dewetting area A versus time t. (b) Nondimensionalized dewetting area A/A_0 versus nondimensionalized time t/τ_i. The inset is a sketch of the dewetting rim.

studies have proposed three physical or mechanical forces that counteract the capillary force to determine the dewetting dynamics at small scales [2,13]. Given that the measured contact line velocity [$\sim O(10^{-1}$ m/s), SM [37], Sec. II] is much greater than that of molecular-kinetics-dominated interfacial motion [43,45,46] [$\sim O(10^{-4}$ m/s)], molecular friction is thus negligible in our dewetting phenomena. To identify hydrodynamic forces governing the fast dewetting, nondimensional analyses were performed using scaling arguments (Appendix B). It turns out that both the characteristic capillary-inertial time $\left[\tau_\mathrm{i} = \sqrt{4\rho R_0^2 h_0/\gamma(\cos\theta_\mathrm{r}^\mathrm{Pho} - \cos\theta_\mathrm{eq}^\mathrm{Pho})}\right]$ and velocity ($V_\mathrm{i} = 2R_0/\tau_\mathrm{i}$) agree with the experimental values on the order of magnitude, where ρ and γ are liquid density and surface tension, respectively, and h_0 is the characteristic rim thickness (SM [37], Sec. IV). Moreover, by replotting Fig. 2(a) in the form of A/A_0 versus t/τ_i, the experimental data of 12 sets of tests collapse into two master curves, one for each surface wettability [Fig. 2(b)]. However, no such agreements and data collapse are achieved when using the viscous scaling for nondimensionalization (SM [37], Sec. V). These results were also obtained from dewetting events on noncircular hydrophobic zones (Appendix B). We therefore conclude that the observed dewetting dynamics is dominated by capillary and inertial forces.

The log-log plot in Fig. 2(b) further implies that liquid dewetting follows a power-law correlation $A/A_0 \sim (t/\tau_\mathrm{i})^\alpha$, with $\alpha \approx 4/3$ for both partially wetting surfaces. To understand this nonlinear dewetting behavior, we propose a hydrodynamic analysis involving parameter scaling. Since the dynamics is dominated by capillary and inertial forces, the governing equation describing the receding liquid rim reads as [47–49]

$$\iiint \rho\left[\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}\right] d\Omega + \iint \gamma \nabla\cdot\mathbf{n}\, dA = 0, \quad (1)$$

where $\boldsymbol{v}$ and $\mathbf{n}$ are the velocity and normal unit vectors at the rim surface, respectively, and $d\Omega$ and dA are the volume and surface elements, respectively. Performing scaling analyses on the two force terms and nondimensionalizing relevant variables in Eq. (1) yield $R^{*2}[(\partial^2 R^*)/(\partial t^{*2})] = C_0$ (Appendix C), where $R^* = \tilde{R}/R_0$ with $\tilde{R}$ being the equivalent radius of the dry patch, $t^* = t/\tau_\mathrm{i}$, and $C_0 = -\gamma\tau_\mathrm{i}^2/\rho R_0^3$. By solving this equation (SM [37], Sec. VI), we found that $R^* \sim t^{*2/3}$ and further

$$A/A_0 \sim R^{*2} \sim t^{*4/3}, \quad (2)$$

which has been experimentally observed [Fig. 2(b)]. It is noted that the capillary-inertial dewetting at length scales of micrometers in this Letter differs from that at length scales of millimeters or larger in previous studies [32,50,51], where liquid contraction progresses with a constant velocity (i.e., $R^* \sim t^*$) described by the Taylor–Culick law [52,53].

We next performed MD simulations to probe the physical landscape of capillary-inertial dewetting at the molecular level. Figure 3(a) illustrates the initial configuration of the simulation system, where a quasi-two-dimensional water nanodrop is symmetrically deposited and pre-equilibrated on a hydrophobic zone with its contact lines pinned on the hydrophilic regions on two sides. The drop consists of 10 304 water molecules, and the substrate is constructed from a face-centered cubic solid. The interactions among water molecules were described using the extended simple point-charge model [54], while the liquid–solid interaction was defined by the widely used Lennard–Jones potential [21,55,56]. By setting different binding energies between liquid and solid molecules, we modeled chemically patterned surfaces with similar wetting properties of the experiments, on which the contact radii of pre-equilibrated nanodrops were in the range of 20.4–21.8 nm. Details about MD simulations are provided in the SM [37], Sec. VII. The drop–surface system in Fig. 3(a) is thermodynamically metastable, and the drop would spontaneously move from the hydrophobic zone to the hydrophilic region.

Using the designed metastable drop–surface configuration, we simulated dewetting processes of water nanodrops from diverse wettable surfaces. In general, we have identified two slightly different dewetting behaviors, depending on the length [L_0, Fig. 3(b)] and wetting property ($\theta_\mathrm{eq}^\mathrm{Pho}$) of the hydrophobic strips. Selected snapshots of two representative cases on hybrid hydrophilic–hydrophobic surfaces with $L_0 = 9.8$ and 16.8 nm whose wetting properties are in accordance with that in Fig. 1(c) ($\theta_\mathrm{eq}^\mathrm{Phi} = 47.4 \pm 0.7°$ and $\theta_\mathrm{eq}^\mathrm{Pho} = 113.9 \pm 0.2°$) are shown in Figs. 3(b) and 3(c). The dewetting processes were quantitatively characterized by tracing contact line positions and measuring contact angles on both sides, and relevant results are shown in Figs. 3(d) and 3(e) and Movie 2. Drop dynamics began with a quasi-stationary scenario (denoting as regime I), wherein the contact line on the right slowly moved leftward under molecular thermal motion, resulting in a slight increase in the contact angle (θ_R), while both the contact line and

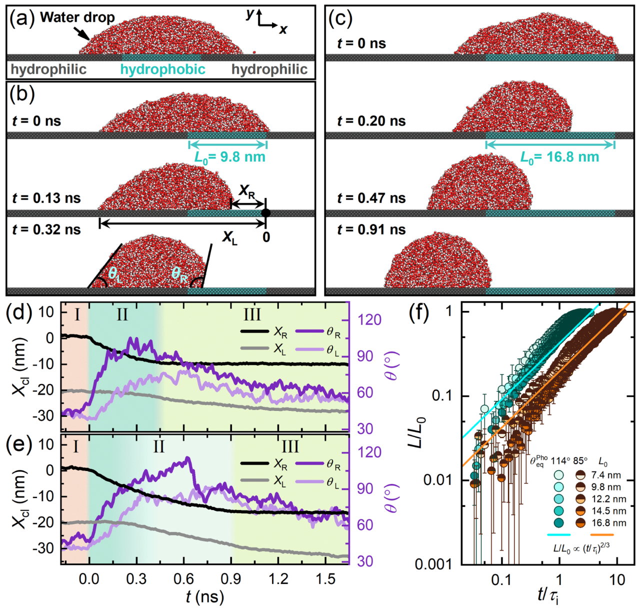

FIG. 3. (a) Metastable drop–surface configuration for MD simulations. Oxygen atoms: red; hydrogen atoms: white; solid atoms in the hydrophobic region: cyan; and solid atoms in the hydrophilic region: gray. (b),(c) Snapshots of a dewetting water drop from a hydrophobic zone with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 114°$ and $L_0 = 9.8$ (b) and 16.8 nm (c). (d),(e) Instantaneous contact line positions X_{cl} and contact angles θ versus time t on hydrophobic zone ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 114°$) with $L_0 = 9.8$ (d) and 16.8 nm (e). (f) Normalized length of the dry patch L/L_0 versus nondimensionalized time t/τ_{i}. The solid lines show $L/L_0 \sim (t/\tau_{\mathrm{i}})^{2/3}$.

contact angle (θ_{L}) on the left remained nearly unchanged. Note that this stochastic thermal-actuated contact line motion occurred with equal probability on either side of the drop (SM [37], Sec. VIII). Once crossing the wettability boundary at $t = 0$ ns, a net capillary force is built up at the contact line, driving the liquid to quickly withdraw from the submerged hydrophobic zone. On hydrophobic surfaces with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 113.9 \pm 0.2°$ and $L_0 \lesssim 9.8$ nm, drop dewetting progressed with the right contact line receding but the left remaining almost at rest, while θ_{L} continuously increased with time, θ_{R} first increased and then decreased after approaching $\theta_{\mathrm{eq}}^{\mathrm{Pho}}$ at $t \approx 0.25$ ns [Fig. 3(d)]. Note that liquid motion was constrained in the hydrophobic zone throughout the dewetting process (regime II), bearing some similarity with the experimental observations [Fig. 1(c)]. Such constrained motion also happened at early times of nanodrop dewetting on other hydrophobic surfaces [e.g., the one with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 113.9 \pm 0.2°$ and $L_0 = 16.8$ nm in Fig. 3(c)]. However, when θ_{L} exceeded the advancing contact angle on the hydrophilic zone $\theta_{\mathrm{a}}^{\mathrm{Phi}}$ [e.g., ~68° at $t \approx 0.2$ ns in Fig. 3(e)], the left contact line started advancing slowly on the hydrophilic region, that is, a slow wetting scenario appeared and persisted for the remaining time of dewetting. After the whole drop moved to the hydrophilic region, the drop–surface system was still not in equilibrium because both θ_{L} and θ_{R} were greater than $\theta_{\mathrm{eq}}^{\mathrm{Phi}}$, giving rise to a relaxation regime (regime III) of solely advancing the left contact line to the end.

The presence of a wetting scenario in the late stage of nanodrop dewetting is due to the accumulation of liquid molecules transported from the receding side to the pinning side, and its occurrence highly depends on $\theta_{\mathrm{a}}^{\mathrm{Phi}}$, $\theta_{\mathrm{r}}^{\mathrm{Pho}}$ and L_0 (Appendix D). Nevertheless, we found that it does not affect the main dynamics of the capillary-inertial dewetting (Appendix D) because of its very low velocity (typically 4–20 times slower than the dewetting velocity). Figure 3(f) reports the temporal evolutions of the growing dry patches (L) in the nondimensional form. Similar to the experimental findings in Fig. 2(b), all simulated data on hydrophobic strips with different lengths ($L_0 = 7.4$–16.8 nm) collapse onto two separate master curves with similar slopes in the log-log plot, corresponding to different surface wettability. More importantly, the dewetting dynamics follows a 2/3 power law [i.e., $L/L_0 \sim (t/\tau_{\mathrm{i}})^{2/3}$], which also agrees with the experimental results, as suggested by Eq. (2). These findings provide strong evidence that the experimentally identified dewetting dynamics have been fairly well reproduced by MD simulations. It should be noted that the capillary-inertial dewetting obeys a power law with a wettability-independent exponent, while the capillary-inertial wetting has been reported to follow a power law with a wettability-dependent exponent (i.e., $R_{\mathrm{w}}^* \sim t^{*\beta}$ with $\beta = 0.2$–0.5, Table S2 [37]) [20,21,24,25], except for the very early times ($t \lesssim 0.1\tau_{\mathrm{i}}$ on hydrophobic surfaces [22,23]).

To understand the dewetting dynamics in detail, molecular motions associated with the moving contact lines were further analyzed. Here, we took the advantage of the coexistence of liquid dewetting and wetting in Fig. 3(e) to compare the kinetic motions of advancing and receding contact lines at the molecular scale through labeling and tracking interfacial water molecules (SM [37], Sec. IX). Figure 4(a) displays molecular motions of contact lines in a short-time interval of 17 ps for the dewetting drop in Fig. 3(c). We found that the left part of the drop spread on the solid via the sliding approach, as suggested by the transverse velocity vectors, agreeing with the literature [14]. Specifically, the excess free energy of the drop–surface system is directly imparted to the spreading liquid in the form of mechanical work, powering the forward movements of all interfacial molecules at the contact line. This sliding motion persists throughout the whole wetting process, which can be inferred by longtime tracking of the mean trajectory of labeled water molecules since the beginning of dewetting, wherein the transverse velocity is always identified [Fig. 4(b)]. Nevertheless, a slightly upward displacement of the tracked molecule clusters [~0.8 nm, Fig. 4(b) and Movie 3] was also caused by the weakened random thermal motion of water molecules on solid surfaces

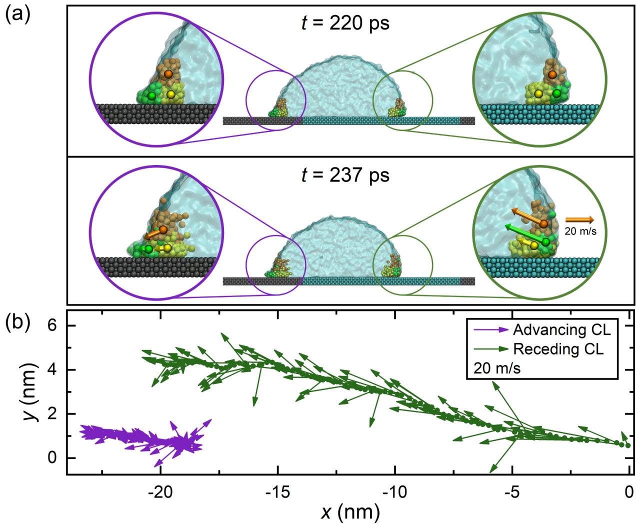

FIG. 4. (a) Enlarged view of the molecular motion within a time interval of 17 ps near the left and right contact lines for the dewetting event in Fig. 3(c). Water nanodrop: transparent; water molecules at the three-phase contact line (CL): green; at the solid–liquid interface: yellow; and at the liquid–gas interface: orange. Solid arrows denote velocity vectors of mass centers. (b) Plot of the spatial locations of the mass center for the marked water molecules at $t = 0$ ns with its instantaneous velocity vector during the liquid dewetting process [i.e., regime II in Fig. 3(e)]. The time interval between two adjacent points is 10 ps.

($\ell_{\mathrm{diff}} \sim \sqrt{D_{\mathrm{w}} t_{\mathrm{D}}} \approx 1.4$ nm), where $D_{\mathrm{w}} = 2.3 \times 10^{-9}$ m^2/s is the self-diffusion coefficient of water molecules at 25 °C and $t_{\mathrm{D}} \approx 0.9$ ns is the diffusion time.

By contrast, the right part of the nanodrop dewetted from the solid through a nontrivial mode: in each elementary movement of the contact line, liquid molecules within the three-phase confluence region (green) desorbed from the solid to minimize the interfacial free energy and then ascended along the drop surface with their neighboring molecules at the liquid–gas interface (orange), while the solid–liquid interface (yellow) evolved as the new receding front [Fig. 4(a)]. Their left-upward velocities imply a rollinglike motion, contrasting with that of the wetting left side. From Fig. 4(b) it is clear that the tracked molecules first moved upward until attaining a maximum displacement of ∼4 nm, which is more than three times that of $\ell_{\mathrm{diff}} \approx 1.17$ nm ($t_{\mathrm{D}} \approx 0.6$ ns), and then directed downward thereafter. Such swirlinglike velocity field further confirms the rolling motion (Movie 3). Similar results have also been obtained from the constrained dewetting events (Appendix E). Notably, rolling contact line motion was also identified in drop spreading—but only at the very beginning stage where the contact angle was close to 180°, while the prevalence of the motion was sliding [14]. Therefore, the motions of liquid molecules during dynamic wetting and dewetting processes are dissimilar.

To summarize, we resolved the spontaneous, capillary-driven dewetting dynamics at the microscopic scale and explicitly demonstrated that the dewetting process is not the reversal of wetting. First, although fast dewetting is still dominated by capillary and inertial forces, the dynamics follows a power law with a *wettability-independent* exponent, which differs from the *wettability-dependent* power-law dynamics of capillary-inertial liquid wetting [20,21,24,25]. Second, molecular motion analyses have revealed that liquid dewetting proceeds via a rolling mode, which is different from sliding-dominated liquid wetting [14].

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End Matter

Appendix A: Dewetting phenomena on hydrophobic zones of noncircular shapes—To demonstrate the generality of our results, we also studied liquid dewetting phenomnona on other eighteen hydrophobic zones with two wetting properties ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$) and three geometric sizes (lengths of $l_0 = 50$ μm, 200 μm, and 400 μm), and shapes (i.e., isosceles acute triangle, square, and parallel hexagon), which have sharp or obtuse corners, as illustrated in Fig. 5(a). It is evident from the illustrative examples in Figs. 5(b)–5(d) that the dewetting phenomena are similar to those on circular hydrophobic zones in Fig. 1(c), and detailed descriptions are provided in the SM [37], Sec. III.

Appendix B: Nondimensional analyses—The capillary force driving the receding liquid can be given by $F_{\mathrm{c}} \sim 2\gamma R_0(\cos\theta_{\mathrm{r}}^{\mathrm{Pho}} - \cos\theta_{\mathrm{eq}}^{\mathrm{Pho}})$ [2,8,13], the inertial force of the moving liquid rim scales as $F_{\mathrm{i}} = m_0 a \sim 8\rho R_0^3 h_0/\tau_{\mathrm{i}}^2$, and the viscous force arising from the laminar shear flow can be estimated by $F_{\mathrm{v}} = \mu A_0 \dot{\varepsilon} \sim 8\mu R_0^3/h_0\tau_{\mathrm{v}}$. Here, τ_{i} and τ_{v} are the characteristic capillary-inertial and capillary-viscous times, respectively; μ is the liquid viscosity; $m_0 \sim 4\rho R_0^2 h_0$ is the mass of the moving rim; the characteristic rim thickness h_0 was respectively estimated to be ~28 μm and ~40 μm on the hydrophobic zones with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and 102° from the confocal microscopy images of water films just before dewetting (SM [37], Sec. IV); $a \sim 2R_0/\tau_{\mathrm{i}}^2$ is the acceleration; and $\dot{\varepsilon} \sim 2R_0/h_0\tau_{\mathrm{v}}$ is the rate of shear deformation. Balancing the driving capillary force with each of the two resistance forces yields $\tau_{\mathrm{i}} = \sqrt{4\rho R_0^2 h_0/\gamma(\cos\theta_{\mathrm{r}}^{\mathrm{Pho}} - \cos\theta_{\mathrm{eq}}^{\mathrm{Pho}})}$ and $\tau_{\mathrm{v}} = 4\mu R_0^2/\gamma(\cos\theta_{\mathrm{r}}^{\mathrm{Pho}} - \cos\theta_{\mathrm{eq}}^{\mathrm{Pho}})h_0$. While the former

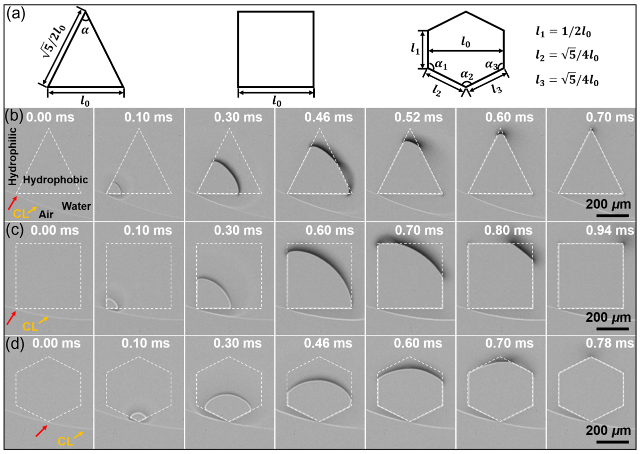

FIG. 5. (a) Sketch of the three shapes of hydrophobic zones with sharp or obtuse corners. From left to right: isosceles acute triangle, square, and parallel hexagon. (b)–(d) Selected snapshots of a contact line moving across hydrophobic zones ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$, denoted by dashed lines) with geometric shapes of an isosceles acute triangle (b), a square (c), and a parallel hexagonal (d). The sizes of these hydrophobic zones are described by their characteristic length $l_0 = 400$ μm, which are defined in (a). The red arrows denote the intersection points of the contact lines with the hydrophobic zones, and the yellow arrows point out the contact lines (CL).

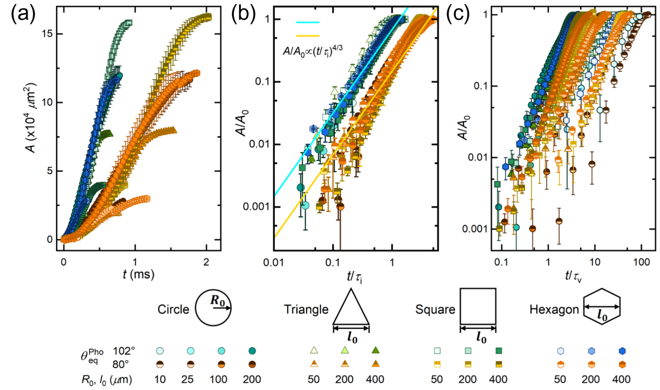

FIG. 6. (a) The dewetting area A versus time t on hydrophobic zones of different shapes (circle; isosceles acute triangle; square; and parallel hexagon), sizes ($R_0 = 10$–200 μm for circular hydrophobic zones, and $l_0 = 50$–400 μm for hydrophobic zones of other shapes) and wetting properties ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and 102°). (b) Nondimensionalized dewetting area A/A_0 versus nondimensionalized time t/τ_{i} for the data in (a). (c) Log-log plot of A/A_0 vs. t/τ_{v}.

predicts a characteristic time $\tau_{\mathrm{i}} \sim O(10^{-2}$–$10^{-1}$ ms) and a characteristic velocity $V_{\mathrm{i}} = 2R_0/\tau_{\mathrm{i}} \sim O(0.1$ m/s), well matching the experimental data (Table S3 [37]), the latter suggests $\tau_{\mathrm{v}} \sim O(10^{-4}$–$10^{-1}$ ms) and $V_{\mathrm{v}} = 2R_0/\tau_{\mathrm{v}} \sim O(1$–$10$ m/s), which are typically 1–2 orders of magnitude different from the experimental values.

Figure 6(a) shows the temporal evolutions of the growing dry area A on hydrophobic zones of different wetting properties, geometric shapes, and sizes. Similar to the observations in Fig. 2(a), A nonlinearly increases with t, and the dewetting process exhibits a strong dependence on the size (A_0) and wetting property ($\theta_{\mathrm{eq}}^{\mathrm{Pho}}$) of the hydrophobic zones. By replotting Fig. 6(a) in terms of $\log(A/A_0)$ vs. $\log(t/\tau_{\mathrm{i}})$, all experimental data collapse onto two linear curves with similar slopes, corresponding to each $\theta_{\mathrm{eq}}^{\mathrm{Pho}}$, as shown in Fig. 6(b). However, no such data collapse were achieved via nondimensionalizing with the viscous scaling (τ_{v}) [Fig. 6(c)]. These findings imply that capillary and inertial forces dominate the dewetting dynamics at the microscopic scale.

Appendix C: Hydrodynamic model of liquid dewetting—Assuming that all liquid withdrawing from the dry patch is collected in the moving rim [30,44,57], the inertial resistance force in Eq. (1) can be estimated as $\iiint \rho\{[(\partial \boldsymbol{v})/(\partial t)] + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}]\} d\Omega \approx \rho \tilde{R}^2 h_0 \ddot{\tilde{R}}$. Moreover, the driving capillary force can be approximated as $\iint \gamma \nabla \cdot \mathbf{n} dA \approx -\gamma[(1/R_{\mathrm{d}}) - (1/\tilde{R})] h_0 \tilde{R} \approx \gamma h_0$, where R_{d} is the curvature radius of the evaporating droplet [Fig. 2(b)] and $R_{\mathrm{d}} \gg \tilde{R}$ is used. Substituting these correlations into and nondimensionalizing the relevant variables in Eq. (1) yield $R^{*2}[(\partial^2 R^*)/(\partial t^{*2})] = C_0$. By solving this equation (SM [37], Sec. VI), we found that $R^* \sim t^{*2/3}$ or equivalently $A/A_0 \sim R^{*2} \sim t^{*4/3}$. It is noteworthy that this model can well describe all experimental data on various hydrophobic surfaces with different wetting properties, geometric shapes, and sizes, as shown in Fig. 6(b) (SM [37], Sec. III). Moreover, our hydrodynamic model was developed based on the same physical principle as that of the Taylor–Culick theory [52,53]—Newton's Second Law, but under different parameter scalings of the capillary and inertial forces.

Appendix D: The occurrence of slow liquid wetting in nanodrop dewetting and its negligible role on the dewetting dynamics—As shown in Fig. 3(e), once liquid dewetting was initiated, the contact angle on the pinned left side θ_{L} started to increase because of the accumulation of liquid molecules that were transported from the receding side. If θ_{L} exceeded the advancing contact angle on the hydrophilic region ($\theta_{\mathrm{a}}^{\mathrm{Phi}}$) or equivalently the leftward interfacial force became larger than the pinning force $F_{\mathrm{p}}^{\mathrm{Phi}} = \gamma(\cos\theta_{\mathrm{eq}}^{\mathrm{Phi}} - \cos\theta_{\mathrm{a}}^{\mathrm{Phi}})$, liquid wetting would be triggered with slowly advancing the left contact line. This nanodrop dewetting accompanying by slow wetting at the late stage has been observed on hydrophobic surfaces with sufficiently large sizes (L_0) or comparatively weaker hydrophobicity ($\theta_{\mathrm{eq}}^{\mathrm{Pho}}$) for the drop–surface system in Fig. 3(a) (consisting of 10 304 water molecules) and a larger simulation system (consisting of 36 015 water molecules) in the Supplemental Material

(SM [37], Sec. VII). It is evident that whether this dewetting scenario appears depends on the amount and rate of liquid molecules transporting from the dewetting side to the pinning side. In principle, one could suppress the wetting scenario in nanodrop dewetting by either increasing θ_a^{Phi} or decreasing θ_r^{Pho} (i.e., tuning the contact angle hysteresis) or shortening L_0 while keeping the other two factors unaltered.

Despite its appearance, the wetting scenario does not affect the main dynamics of the capillary-inertial dewetting. First, we found that both the durations and velocities of dewetting events involving slow wetting still fairly agree with the characteristic capillary-inertial values on the order of magnitude [e.g., $t_{\mathrm{dw}} \approx 1.0$ ns and $v_{\mathrm{cl-dw}} \approx L_0/t_{\mathrm{dw}} \approx 17$ m/s while $\tau_{\mathrm{i}} \approx 0.3$ ns and $V_{\mathrm{i}} \approx 56$ m/s for the case in Fig. 3(e)]. Second, by performing nondimensional analyses on the temporal evolutions of the dewetting radii of dry patches (L), all simulation data on hydrophobic strips of different lengths ($L_0 = 7.4$–16.8 nm) collapse onto two linear curves with similar slopes for each surface wettability in the log-log plot, regardless of whether there is a wetting scenario, as shown in Fig. 3(f) and SM [37], Sec. VII. Third, all detwetting events in the MD simulations obey the same power law, $L/L_0 \sim (t/\tau_{\mathrm{i}})^{2/3}$, as identified in the experiments and suggested by the proposed hydrodynamic model [i.e., Eq. (2)].

Appendix E: Molecular motions in constrained dewetting processes—To illustrate the molecular kinetics in the constrained dewetting events, we tracked the molecular motion near the left pinned and right receding contact lines during the dewetting of a water nanodrop on a hydrophobic surface with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 113.9 \pm 0.2°$ and $L_0 = 9.8$ nm in Fig. 3(b). As shown in Fig. 7, a swirlinglike velocity field, which indicates a superposition of a translation with respect to the surface from the right to the left and a counterclockwise rotation around its own axis, has also been obtained for the tracked interfacial molecules on the receding (right) side, contrasting to the thermally-induced weak diffusion on the pinned (left) side. That is, liquid dewetting proceeds with a rolling motion as that in Fig. 4(b).

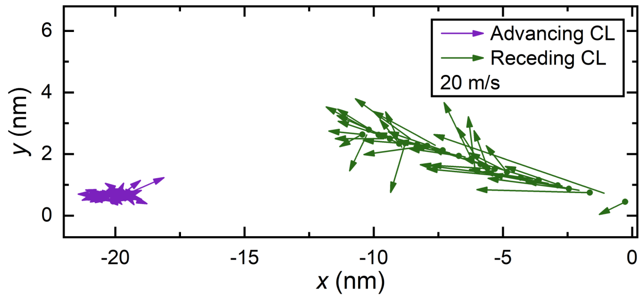

FIG. 7. Plot of the spatial locations of the mass center for the marked water molecules at $t = 0$ ns with its instantaneous velocity vector during the liquid dewetting process [i.e., regime II in Fig. 3(d)] on a hydrophobic surface with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 113.9 \pm 0.2°$ and $L_0 = 9.8$ nm. The time interval between two adjacent points is 10 ps.

Supplemental Material for

Spontaneous Capillary-Inertial Dewetting at the Microscopic Scale

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CONTENTS

Supplementary Section I. Experimental Methods

Sample preparation

Thin glass slides as base surfaces were first immersed in a mixture of sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2) at a volume ratio of 7:3 for $8\ \mathrm{h}$ at $25\ ℃$ and then rinsed with deionized water and dried with nitrogen, resulting in clean and homogeneous hydrophilic surfaces. The standard photolithography method was then employed to fabricate hydrophobic patterns on glass plates. With the help of various masks, the hydrophobic patterns were defined by an HTI 751 photoresist film (of thickness $\sim 0.6\ \mu\mathrm{m}$), which were spin coated on the baked glasses first at $600\ \mathrm{rpm}$ for $6\ \mathrm{s}$ and then at $3000\ \mathrm{rpm}$ for $40\ \mathrm{s}$. Afterwards, the glasses were softly baked at $90\ ℃$ for $2\ \mathrm{min}$ and then cooled to room temperature. Subsequently, they were exposed to ultraviolet (UV) light at a wavelength of $365\ \mathrm{nm}$ and a power of $9\ \mathrm{mW/cm^2}$ for $5\ \mathrm{s}$ via masks with circle arrays. With postexposure baking at $100\ ℃$ for $2\ \mathrm{min}$ and then being cooled to room temperature, the glasses were immersed in the HTI Developer for $1\ \mathrm{min}$, rinsed with deionized water and blown dry with nitrogen. After plasma cleaning at $200\ \mathrm{W}$ for $3\ \mathrm{min}$, the as-prepared surfaces were placed in a desiccator with $4\ \mu\mathrm{L}$ 1H,1H,2H,2H-perfluorodecyl-triethoxysilane (97%; Sigma–Aldrich, USA), and then placed in an oven at $80\ ℃$ for $10\ \mathrm{h}$ and $85\ \mathrm{min}$, respectively, rendering surfaces with varying hydrophobicity. Finally, the exhausted photoresists were removed using acetone, isopropanol, and ethanol under ultrasonic conditions for $20\ \mathrm{min}$, and the as-prepared surfaces were rinsed with deionized water and dried with nitrogen for subsequent dewetting experiments.

Surface characterization

Twelve hybrid hydrophilic–hydrophobic surfaces were fabricated through a standard photolithography combined with selected silanization as described above. The corresponding hydrophobic-to-hydrophilic area ratios [$\varphi = \pi R_0^2/[(2R_0 + S)^2 - \pi R_0^2]$ with R_0 and S being the radii of the circular hydrophobic zones and their spacing, respectively, **Fig. S1**(**a**)] are less than 0.1. It can be predicted theoretically by the well-known Cassie–Baxter law that such low φ would not significantly affect the overall surface wettability [1]. To experimentally confirm this point, we conducted contact angle measurements on the homogeneous hydrophilic and hydrophobic surfaces (i.e., pure glass surfaces and hydrophobized glass surfaces by 1H,1H,2H,2H-perfluorodecyl-triethoxysilane) and all the hybrid surfaces using a commercial goniometer (OCA25, Dataphysics, Germany). As summarized in **Table S1**, the equilibrium, advancing and receding contact angles on hybrid hydrophilic–hydrophobic surfaces are close to those on the hydrophilic surfaces. The roughness of the hybrid hydrophilic–hydrophobic surfaces was characterized by a high-resolution atomic force microscope (BioScope Resolve™, Brucker, USA), and a representative topographic image is displayed in **Fig. S1**(**b**). The root-squared roughness of the hydrophilic and hydrophobic zones are $\sim 0.60\ \mathrm{nm}$ and $\sim 0.33\ \mathrm{nm}$, respectively. The cross-section profile in **Fig. S1**(**c**) shows that the height difference between the hydrophilic and hydrophobic zones is about $2\ \mathrm{nm}$, which is four orders of magnitude smaller than the characteristic height of the dewetting water film ($h_0 = 40\ \mu\mathrm{m}$). Such a small roughness difference at the wettability boundary would not affect the dewetting process in a notable manner and therefore the influence of surface geometry on the dewetting dynamics can be reasonably excluded.

Dewetting experiments

Millimetre-sized drops of pure water ($18.4\ \mathrm{M\Omega \cdot cm}$, Millipore Synergy, Darmstadt, Germany) were deposited on the as-prepared surfaces from a syringe needle. The drop dewetting was triggered by its spontaneous evaporation at the room temperature $\mathrm{T} = 20 \pm 1\ ^{\circ}\mathrm{C}$, ambient pressure $P = 1\ \mathrm{atm}$ and relative humidity $\mathrm{RH} = 60 \pm 1\%$. The retraction of the three-phase contact line during the dewetting process was observed from the bottom by an inverted microscope (Nikon, Japan) and recorded by a high-speed camera (Phantom, V2012, USA) at $30{,}000 - 16{,}0000\ \mathrm{fps}$. Prior to dewetting, the three-dimensional profile near the contact line of the water drop was scanned by a confocal microscopy (Nikon A1 10X dry objective, Japan) from the bottom to the top at a step of $15\ \mu\mathrm{m}$.

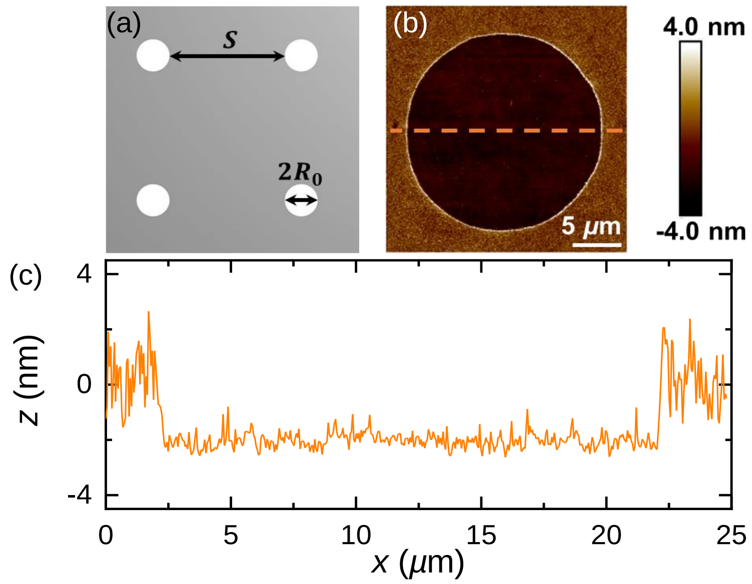

Fig. S1. Surface characterizations. (a) Schematic diagram of the hybrid hydrophilic–hydrophobic surface, R_0 and S are the radius of the hydrophobic zone and their spacing, respectively. (b) AFM image of a hydrophilic glass surface patterned with a hydrophobized zone with $R_0 = 10\ \mu\mathrm{m}$. (c) The cross-section profile of the hybrid surface along the dashed orange line in (b).

Supplementary Section II. Dewetting time and dewetting velocity on circular hydrophobic zones

During dewetting, the contact line fast recedes with the shape of a circular, and a representative one with $R_0 = 150\ \mu\mathrm{m}$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$ is shown in **Fig. S2**. The dewetting time t_{dw}, which is defined as the total duration of the moving contact line on the hydrophobic zone, is one of the most important characteristic parameters of the dewetting process. To rule out the effect of the evaporation-driven dewetting on the subsequent capillary-driven dewetting, we scrutinized the dewetting time under different evaporating rates on diverse hybrid surfaces with different "hydrophobic" zones with six sizes ($R_0 = 10\ \mu\mathrm{m}$, $R_0 = 25\ \mu\mathrm{m}$, $R_0 = 50\ \mu\mathrm{m}$, $R_0 = 100\ \mu\mathrm{m}$, $R_0 = 150\ \mu\mathrm{m}$, $R_0 = 200\ \mu\mathrm{m}$) and two wetting properties ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and $102°$), and relevant results are summarized in **Figs. S3**(**a**) and **S3**(**b**). Although the data is scattered, it is obvious that on any given hybrid surface, i.e., a specified $\theta_{\mathrm{eq}}^{\mathrm{Pho}}$ and R_0, t_{dw} keeps nearly constant in the range of evaporation-driven dewetting velocity $\bar{v}_{\mathrm{cl-ev}}$ of $0.3-19\ \mu\mathrm{m/s}$. Meanwhile, with the increase of R_0 from 10 to $200\ \mu\mathrm{m}$, t_{dw} linearly increases from 0.04 to 0.95 ms for $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$, and from 0.11 to 2.20 ms for $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$, respectively. Furthermore, the calculated dewetting velocities $v_{\mathrm{cl-dw}} = 2R_0/t_{\mathrm{dw}}$ also exhibit no dependence on the size of the hydrophobic zone, and was measured to be $\sim 0.5\ \mathrm{m/s}$ and $\sim 0.2\ \mathrm{m/s}$ on "hydrophobic" zones with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$ and $80°$, respectively. Since $v_{\mathrm{cl-dw}}$ is about six orders of magnitude higher than $\bar{v}_{\mathrm{cl-ev}}$, we conclude that the initial contact line motion driven by evaporation cannot affect the subsequent spontaneous, capillary-driven dewetting processes.

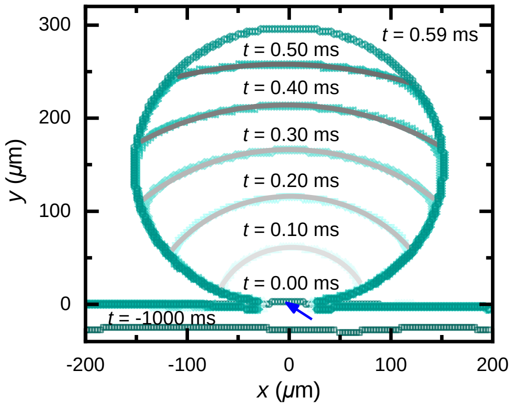

Fig. S2. Time-lapse profiles of the receding contact line in **Fig. 1(c)**. The blue arrow denotes the intersection point of the contact line with the hydrophobic circular zone. The gray solid lines are circular fits of the experimental data.

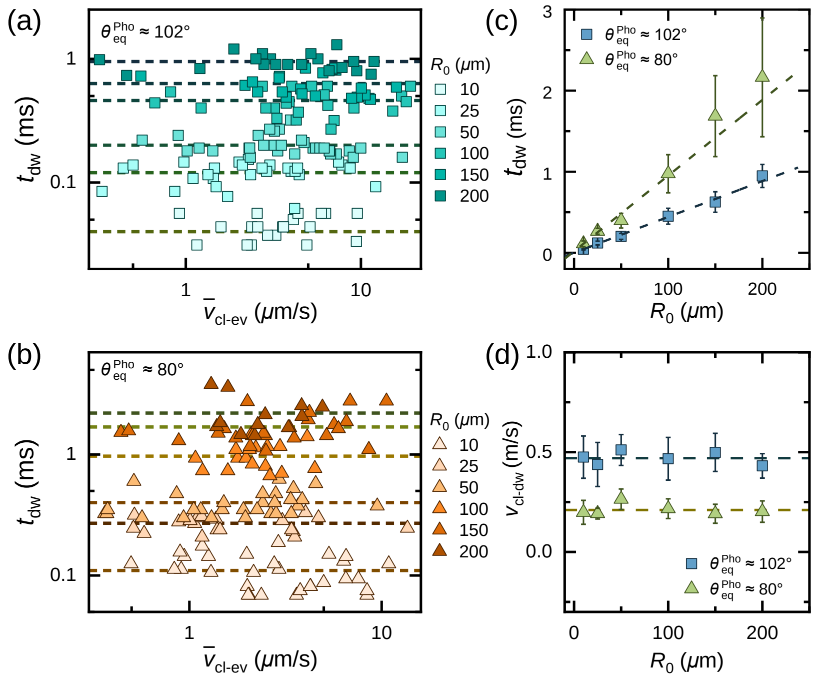

Fig. S3. Characterization of the dewetting processes. Plot of the dewetting time t_{dw} on the hybrid surfaces of (a) $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$, and (b) $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$, as a function of the average velocity of the moving contact line induced by evaporation $\bar{v}_{\mathrm{cl-ev}}$. The dashed lines show the mean value of t_{dw}. (c) Plot of t_{dw} versus the radius of the circular hydrophobic zone R_0. The dashed lines show best linear fits. (d) Plot of the estimated dewetting velocity $v_{\mathrm{cl-dw}}$ as a function of R_0. The dash lines are guides for eyes.

Supplementary Section III. Dewetting phenomena on other hybrid hydrophobic–hydrophilic surfaces

Dewetting phenomena on other circular hydrophobic zones

We have conducted dewetting experiments on hydrophilic surfaces patterned with circular hydrophobic zones of six different sizes ($R_0 = 10 - 200\ \mu\mathrm{m}$) and two wetting properties ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and $102°$). In the main text, a representative dewetting phenomenon on the hybrid surface with $R_0 = 150\ \mu\mathrm{m}$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$ has been shown in **Fig. 1(c)**. Here, we further provide experimental data on circular "hydrophobic" zones with three different radii ($R_0 = 10\ \mu\mathrm{m},\ 50\ \mu\mathrm{m}$ and $200\ \mu\mathrm{m}$) and two different wetting properties ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and $102°$) in **Fig. S4**. From the selected snapshots of receding contact lines travelling across different hydrophobic zones, we can clearly identify similar dewetting scenarios as discussed in the main text: starting with the evaporation-driven slow dewetting and following by fast capillary-driven dewetting, which proceeds with a self-similar contact line motion.

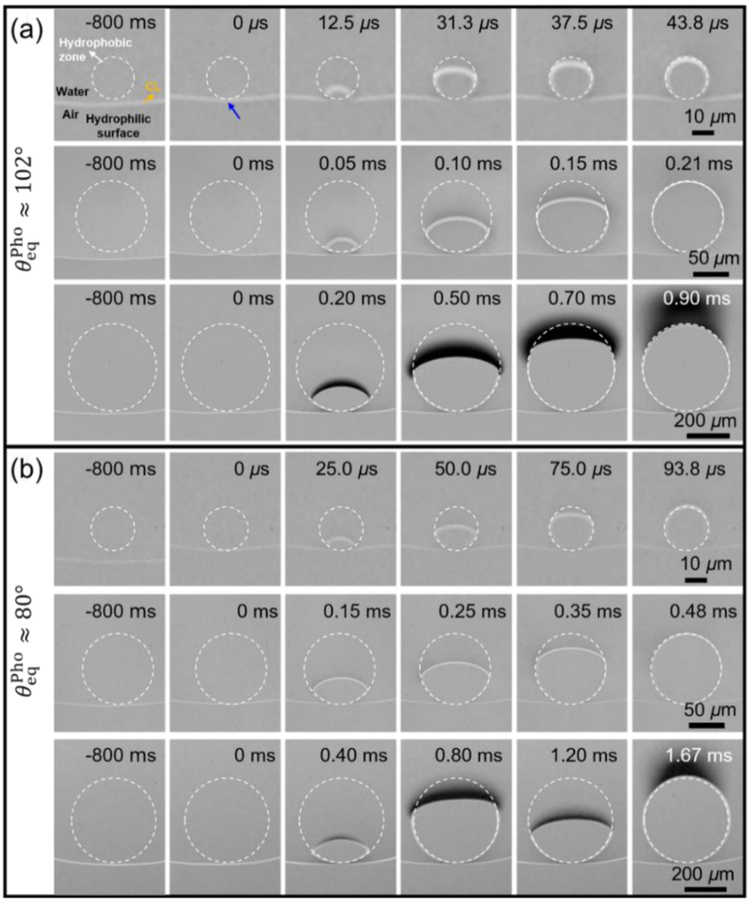

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Fig. S4. Dewetting phenomena on various hybrid surfaces. Selected snapshots of a contact line moving across a circular hydrophobic zone with (a) $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$, $R_0 =$

$10\,\mu\mathrm{m}$, $50\,\mu\mathrm{m}$, $200\,\mu\mathrm{m}$ (from top to bottom), (b) $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$, $R_0 = 10\,\mu\mathrm{m}$, $50\,\mu\mathrm{m}$, $200\,\mu\mathrm{m}$ (from top to bottom). The white dashed lines indicate the boundaries of hydrophobic zones. The starting time ($t = 0\,\mathrm{ms}$) is defined as the moment that the moving contact line touches the wettability boundary.

Dewetting phenomena on hydrophobic zones of non-circular shapes

To check the generality of our results, dewetting experiments were also carried out on hydrophilic glass surfaces patterned with hydrophobic zones of non-circular shapes. As illustrated in **Fig. 5(a)** of the main text, the three types of triangle-shaped hydrophobic zones own the same vertex angle of $\alpha = 53.2°$, while their base lengths l_0 are $50\,\mu\mathrm{m}$, $200\,\mu\mathrm{m}$ and $400\,\mu\mathrm{m}$, respectively. Three types of square-shaped hydrophobic zones, with side lengths l_0 of $50\,\mu\mathrm{m}$, $200\,\mu\mathrm{m}$ and $400\,\mu\mathrm{m}$, were also designed and fabricated on hydrophilic glass surfaces [**Fig. 5(a)**]. For hydrophobic zones in the shape of parallel hexagon, the length ratio of three adjacent edges is $l_1 : l_2 : l_3 = 1 : \sqrt{5}/2 : \sqrt{5}/2$. Here, we have $l_1 = 0.5 l_0$, and the sizes of these three types of hydrophobic zones were set by $l_0 = 50\,\mu\mathrm{m}$, $200\,\mu\mathrm{m}$ and $400\,\mu\mathrm{m}$; the angles between two adjacent edges were $\alpha_1 = 116.6°$, $\alpha_2 \approx 126.8°$ and $\alpha_3 = 116.6°$, as displayed in **Fig. 5(a)**.

In **Figs. 5(b)–5(d)**, we show selected snapshots of the dewetting processes of water films on three hydrophobic zones of different shapes with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$ and $l_0 = 400\,\mu\mathrm{m}$. Similar to the dynamic phenomena on circular hydrophobic zones [see **Fig. 1(c)** in the main text and **Fig. S4** above], once the fast dewetting was triggered, i.e., when the moving contact line passed through the hydrophilic–hydrophobic boundary, a dry patch would immediately emerge and grow radially inwards into the thin film under the capillary force. Obviously, the receding contact line always takes a shape of a circular arc, thanks to the small size of the dewetting area ($l_0 = 50 - 400\,\mu\mathrm{m}$) comparing to the capillary length $l_{\mathrm{c}} = \sqrt{\gamma/\rho g} \sim 2.7\,\mathrm{mm}$, where γ and ρ are the surface tension and density of the liquid, respectively; g is the gravitational acceleration. This finding can be more clearly revealed by superimposing the contours of moving contact lines at different times, which are well fitted by circles, as shown in **Figs. S5(a)–S5(c)**. Moreover, the whole dewetting process was also constrained in the hydrophobic zone and the dewetting time t_{dw} showed no notable dependence on the velocity of the moving contact line prior to dewetting (i.e., $\bar{v}_{\mathrm{cl-ev}}$), but it did highly depend on the size of the hydrophobic zone, as reported in **Fig. S6**.

The temporal evolution of the growing dry area A on eighteen hydrophobic zones of two wetting properties and three different shapes and sizes were further analyzed at the level of the scaling argument. As shown in **Fig. 6(a)**, by normalizing the growing dry area A using the area of the hydrophobic zone A_0, and by non-dimensionalizing the dewetting time t using the capillary-inertial characteristic time $\tau_{\mathrm{i}} = \sqrt{4\rho \tilde{R}_0^2 h_0 / \gamma\left(\cos\theta_{\mathrm{r}} - \cos\theta_{\mathrm{eq}}^{\mathrm{Pho}}\right)}$, all experimental data on hydrophobic zones of different shapes (i.e., circular, isosceles acute triangle, square and parallel hexagon) and sizes ($R_0 = 10 - 200\ \mu\mathrm{m}$ for circular hydrophobic zones and $l_0 = 50 - 400\ \mu\mathrm{m}$ for hydrophobic zones of non-circular shapes) are collapsed onto two master curves, which correspond to two surface wetting properties, but follow a same power-law correlation $A/A_0 \sim (t/\tau_{\mathrm{i}})^{4/3}$ [**Fig. 6(b)**]. Here, $\tilde{R}_0 = \sqrt{A_0/\pi}$ is the equivalent circular radius of the hydrophobic zone, h_0 is the characteristic thickness of the dewetting water films, and θ_r denotes the receding contact angle on the hydrophobic zones. We also performed the dimensional analysis for the experimental data in **Fig. 6(a)** using the characteristic capillary-viscous time $\tau_v = 4\mu\tilde{R}_0^2/\gamma(\cos\theta_r - \cos\theta_{eq}^{Pho})h_0$, and relevant results are plotted in **Fig. 6(c).** Evidently, no data collapse was observed, suggesting that the dewetting dynamics here are also predominated by the capillary and inertial forces, while the contribution from the viscous force is negligible. Based on these analyses, we conclude that the dewetting phenomena and thus the underlying dynamics are not dependent on the shape of the hydrophobic zones.

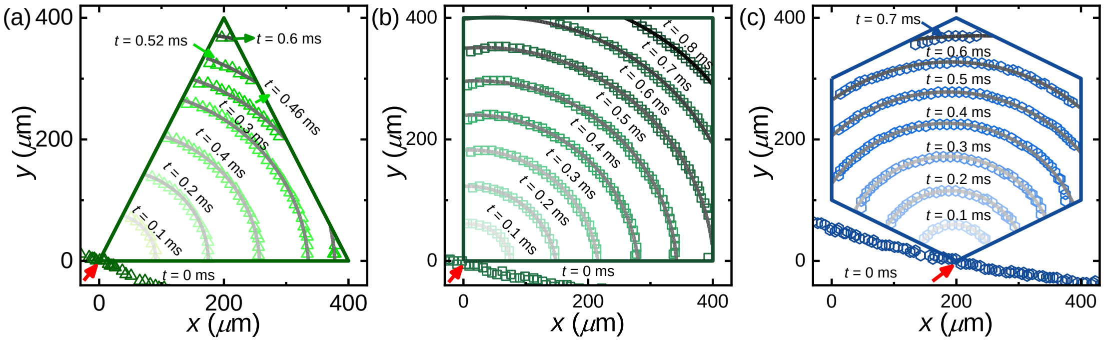

Fig. S5. Time-lapse profiles of the receding contact line of the dewetting water films on hydrophobic zones of an isosceles acute triangle (a), a square (b) and a parallel hexagonal (c) in **Figs. 5(b)–5(d)**. The red arrow denotes the intersection point of the contact line with the hydrophobic zone. The gray solid lines are circular fits of the experimental data.

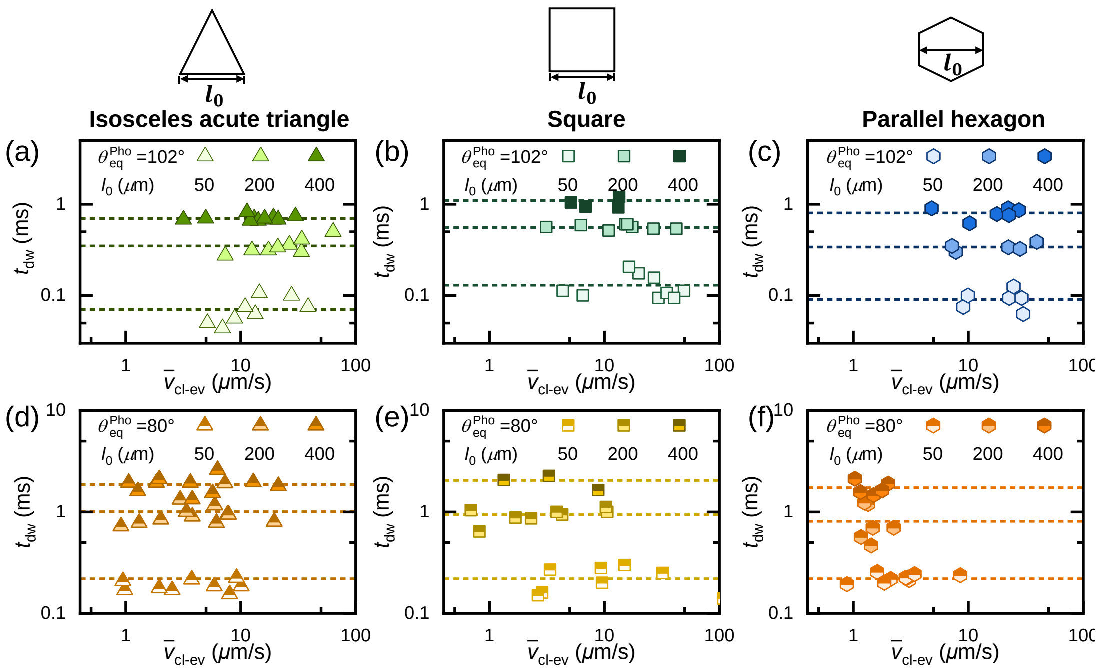

Fig. S6. Plot of the dewetting time t_{dw} on hydrophobic zones of an isosceles acute triangle (a), a square (b) and a parallel hexagonal (c) with $\theta_{eq}^{Pho} \approx 102°$ and with $\theta_{eq}^{Pho} \approx 80°$ as a function of the average velocity of the moving contact line induced by evaporation $\bar{v}_{cl-ev}$. t_{dw} on hydrophobic zones of an isosceles acute triangle (d), a square (e) and a parallel hexagonal (f) with $\theta_{eq}^{Pho} \approx 80°$ vresus $\bar{v}_{cl-ev}$.

The dash lines are guides for eyes.

Supplementary Section IV. Characteristic heights of the dewetting rims

Since the capillary-inertial dewetting proceeds with a duration of $\sim 1\ \mathrm{ms}$ [see **Fig. S3(a)** and **Fig. S3(b)**] and a velocity of $v_{\mathrm{cl-dw}} \sim O(\mathrm{m/s})$ [see **Fig. S3(c)** and **Fig. S3(d)**], it is thus very difficult to directly measure the profile of the receding water film, and determine its characteristic thickness h_0. However, because the hydrophobic zones are rather small, the profiles of the water films prior to dewetting and during dewetting should be on the same order of magnitude. Therefore, we have managed to measure the three-dimensional profiles of receding water films just before the contact line moves across the hydrophobic–hydrophilic boundary, from which h_0 was estimated. It should be noted that patterning hydrophilic surfaces with hydrophobic zones does not cause notable alternation on the wetting properties of water on them, which is evidenced by the very similar values of the measured equilibrium (θ_{eq}), advancing (θ_{a}) and receding (θ_{r}) contact angles on diverse hybrid hydrophobic–hydrophilic surfaces, as summarized in **Table S1**. As a result, receding water films on different hybrid hydrophobic–hydrophilic surfaces should own similar profiles and characteristic thicknesses.

Practically, we performed experimental measurements of water film profiles on the hybrid surfaces with the largest circular hydrophobic zones (i.e., $R_0 = 200\ \mu\mathrm{m}$) using a laser scanning confocal microscopy (Nikon A1 10X dry objective, Japan). The spatial resolution was $\sim 2.47\ \mu\mathrm{m/px}$ in the lateral dimension, while the scanning resolution was $10\ \mu\mathrm{m/step}$ in the vertical direction. The obtained results on the hybrid surfaces with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ are shown in **Fig. 1**(**b**) of the main text and in **Fig. S7** below, respectively. We have chosen the heights of the water films at the center of the hydrophobic zones as the characteristic thicknesses, which were determined to be $\sim 28\ \mu\mathrm{m}$ and $\sim 40\ \mu\mathrm{m}$ for hybrid surfaces with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$ and $102°$, respectively. The rationality of these two characteristic thicknesses was confirmed by the good agreement of the calculated characteristic capillary-inertial time (τ_{i}) and the characteristic capillary-inertial velocity (V_{i}) with the experimental values (t_{dw} and $v_{\mathrm{cl-dw}}$) in the dewetting experiments, as described in the main text. To reflect this more explicitly, we further plotted $t_{\mathrm{dw}}/\tau_{\mathrm{i}}$ and $v_{\mathrm{cl-dw}}/V_{\mathrm{i}}$ as a function of the radius of the hydrophobic zone (i.e., R_0 for circular hydrophobic zones in **Fig. 1c** and $\tilde{R}_0$ for hydrophobic zones of other shapes in **Fig. 5**) in **Figs. S8(a)–S8(b)**. Evidently, both $t_{\mathrm{dw}}/\tau_{\mathrm{i}}$ and $v_{\mathrm{cl-dw}}/V_{\mathrm{i}}$ are on the order of unit, and the experimental data obtained from hydrophobic zones of different shapes (including circle, isosceles acute triangle, square and parallel hexagon) and sizes ($R_0 = 10 - 200\ \mu\mathrm{m}$ for circular hydrophobic zones and $l_0 = 50 - 400\ \mu\mathrm{m}$ for hydrophobic zones of non-circular shapes) collapse onto two curves of different wetting properties, which are independent on R_0 and $\tilde{R}_0$. Furthermore, employing the determined h_0, we have also successfully collapsed the experimental data of the temporal evolution of the growing dewetting area on diverse "hydrophobic" zones onto two master curves [see **Fig. 6(b)**], which follow the same power-law $A/A_0 \sim (t/\tau_{\mathrm{i}})^{4/3}$, as predicted by the developed hydrodynamic model (see **Eq. 2** in the main text).

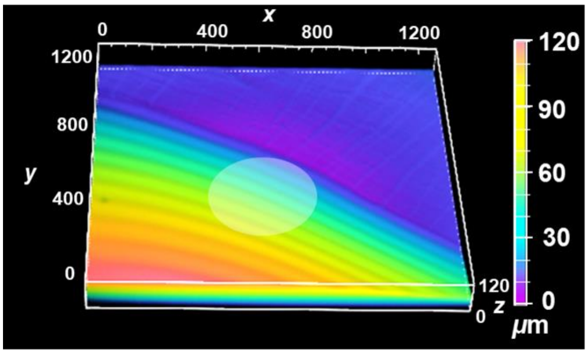

Fig. S7. Characteristic height of the dewetting water film. Three-dimensional confocal microscopy image of a water film just before dewetting on the hybrid hydrophilic–hydrophobic surface with $R_0 = 200\ \mu\mathrm{m}$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 80°$. The white circle represents the submerged hydrophobic zone.

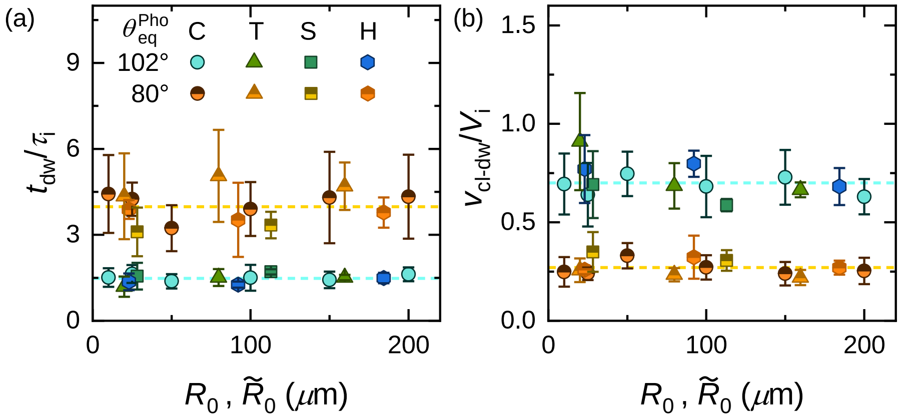

Fig. S8. Variations of the dewetting time t_{dw} normalized by the capillary-inertial characteristic time τ_{i} (a) and dewetting velocity $v_{\mathrm{cl-dw}}$ normalized by the capillary-inertial characteristic velocity V_{i} (b), as a function of the size of the hydrophobic zone (R_0 for circular hydrophobic zones and $\tilde{R}_0$ for hydrophobic zones of other shapes in **Fig. 5**). C: circle; T: isosceles acute triangle; S: square; H: parallel hexagon.

Supplementary Section V. Nondimensional analysis with the characteristic capillary-viscous time

The evolution of the dewetting area A as a function of time t from 12 sets of tests in **Fig. 2**(**a**) of the main text has been further analyzed by non-dimensionalizing using the characteristic capillary-viscous time $\tau_{\mathrm{v}} = 4\mu R_0^2/\gamma(\cos\theta_{\mathrm{r}} - \cos\theta_{\mathrm{e}})h_0$. As shown in **Fig. S9**, no data collapse has been observed, suggesting that the dewetting process is not dominated by liquid viscosity.

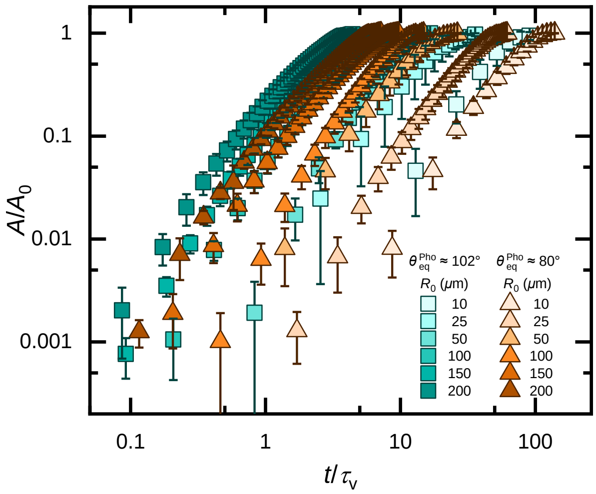

Fig. S9. Nondimensional analysis with the characteristic capillary-viscous time. Plot of the nondimensional dewetting area A/A_0 as a function of the time normalized by the characteristic capillary-viscous time t/τ_{v} for diverse hybrid hydrophobic–hydrophilic surfaces.

Supplementary Section VI. Solving the momentum equation

In the main text, we have attained a simplified Navier–Stokes equation, which reads,

$$R^{*2}\frac{\partial^2 R^*}{\partial t^{*2}} = C_0, \tag{S1}$$

where $R^* = \tilde{R}/R_0$ is the normalized equivalent radius of the dry patch and $C_0 = -\gamma\tau_{\mathrm{i}}^2/\rho R_0^3$. By introducing a temporal parameter $V(R^*) = \frac{\partial R^*}{\partial t^*}$, **Eq. S1** can be rewritten as

$$R^{*2}V\frac{\partial V}{\partial R^*} \sim C_0. \tag{S2}$$

Integrating **Eq. S2** yields

$$\int_{V_1}^{V} VdV \sim \int_{R_1^*}^{R^*} \frac{C_0}{R^{*2}} dR^*, \tag{S3}$$

where the subscript 1 represents the moment before the dewetting process starts. Since $V_1 \sim \bar{v}_{\mathrm{cl-ev}} \ll V$ and $R_1^* \to \infty$, **Eq. S3** can be further simplified as

$$V \sim \sqrt{\frac{-2C_0}{R^*}}. \tag{S4}$$

Substituting **Eq. S4** into the temporal parameter $V(R^*) = \frac{\partial R^*}{\partial t^*}$, and performing the re-integration yields

$$R^* \sim \sqrt[3]{\frac{-9C_0}{2}}\, t^{*2/3}. \tag{S5}$$

It is clear shown from **Eq. S5** that the normalized equivalent radius of the dry patch would increase following a power law relation, regardless of the surface wettability $\theta_{\mathrm{eq}}^{\mathrm{Pho}}$ and the size of the hydrophobic zone R_0, matching well with all experimental and MD simulation results in **Figs. 2(b)**, **6(b)** and **3(f)**.

Supplementary Section VII. Details about the molecular dynamics simulations

General settings of molecular dynamics simulations

To uncover the physical picture of dewetting, we performed molecular dynamics (MD) simulations. A droplet consisting of $10{,}304$ water molecules was symmetrically deposited on a hybrid solid surface formed by a face-centred cubic solid with the (111) surface and a lattice constant of 3.92 Å, as depicted in **Fig. 3**(**a**). Water molecules were simulated using the SPC/E water model [2], which has been extensively validated to describe water properties [3, 4], and the bond length and angle of water molecules were controlled via the SHAKE algorithm. The forces between water molecules and the solid surface were calculated by the Lennard–Jones potential, $U_{\mathrm{LJ}} = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$, where ε and σ denote the binding energy and the collision diameter, respectively, and r is the interatomic distance. ε was varied from 0.2 to $0.4\ \mathrm{kcal/mol}$ to achieve various equilibrium contact angles (**Fig. S10**). The system was first equilibrated in a canonical (*N*, *V*, *T*) ensemble, and the water droplet was moved to the hydrophilic–hydrophobic boundary in a quasistatic path. The subsequent production MD was run for at least 2 ns in a canonical ensemble with a time step of 1.0 fs. The solid atoms were fixed at their equilibrium positions to reduce the computational cost. The long-range Coulombic interactions were computed by the particle–particle particle–mesh (PPPM) method. The system temperature was maintained at the room temperature ($T = 300$ K) by a Nosé–Hoover thermostat via modulating the thermal velocities of water molecules. Periodic boundary conditions were set in all the three directions, and the dimensions in the *x*, *y* and *z* directions were 43.9, 17.2 and 4.7 nm, respectively. The length in the *y* direction is approximately three times the maximum droplet height, and the length in the *x* direction is approximately twice the maximum contact diameter. Such dimensions can avoid unnecessary atom-mirror interactions. All the MD simulations in this work were performed by using the LAMMPS package [5], and all the statistical results were obtained from at least three independent simulations.

In our MD simulations, we controlled the wetting property of solid surfaces by adjusting the binding energy ε for the interatomic interactions between oxygen atoms in water molecules and solid atoms. To determine the equilibrium contact angle of a water nanodrop with a given ε, a molecular system, wherein a quasi-two-dimensional cylindrical water nanodrop of radius $\sim 5\ \mathrm{nm}$ initially sits on the homogeneous solid surface, was first built up [**Fig. S10**(**a**)]; then, independent equilibrium molecular dynamics simulations were conducted in a canonical ensemble ($T = 300\ \mathrm{K}$) for $5\ \mathrm{ns}$ to obtain the equilibrium molecular configurations. The final equilibrium states of water nanodrops on various solid surfaces (different ε) are shown in **Figs. S10(b)–S10(d)**. We have identified the liquid–gas interface by locating the contour profile at the mass density $\rho = 0.5$, i.e., the arithmetical mean of its density in liquid and gas phases [3]. The water contact angle was obtained by fitting the identified liquid gas interface with a truncated circle (**Fig. S10**). In this work, by setting ε as $0.2\ \mathrm{kcal/mol}$, $0.3\ \mathrm{kcal/mol}$ and $0.4\ \mathrm{kcal/mol}$, water contact angles were determined to be $113.9 \pm 0.2°$, $85.1 \pm 0.2°$, and $47.4 \pm 0.7°$, respectively.

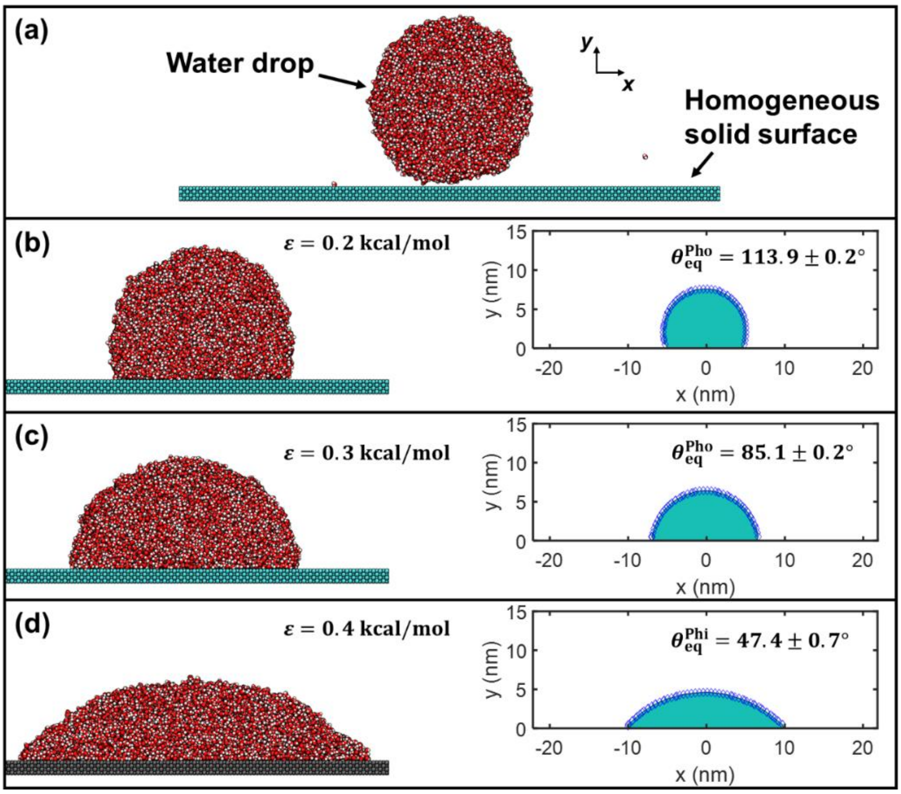

Fig. S10. Calibrating the surface wettability in molecular dynamics simulations. (a) Molecular simulation system for calibrating the surface wettability and its final equilibrium states at (b) $\varepsilon = 0.2$ kcal/mol, (c) $\varepsilon = 0.3$ kcal/mol, (d) $\varepsilon = 0.4$ kcal/mol. The left panels in (b)–(d) show the equilibrium configurations of the molecular systems, while the right panels show the equal-density profiles, in which the liquid phase is colored cyan and the liquid–gas interface is marked by blue diamonds, which are used for fitting with truncated circles.

MD simulations on a larger drop–surface system

We also performed MD simulations of nanodrops consisting of $36{,}015$ water molecules (~ 3.5 times of nanodrop in **Fig. 3a**) on hydrophobic surfaces with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 113.9 \pm 0.2°$ and $L_0 = 16.8 - 34.9$ nm [**Fig. S11(a)**]. Similar to those findings of the drop–surface system in **Fig. 3**, two slightly different dewetting phenomena have been identified, which depends on L_0 [**Figs. S11(b)–S11(c)**]. During whole dewetting process of a nanodrop on the smallest hydrophobic zone with $L_0 = 16.8$ nm, the right contact line solely receded while the left contact line remained almost fixed [**Fig. S11(d)**], analogous to the dewetting behaviour in the experiments [**Fig. 1(c)**]. By contrast, on hydrophobic zones with $L_0 \gtrsim 26.2$ nm, the slow wetting scenario was observed when θ_{L} exceeded $\theta_{\mathrm{a}}^{\mathrm{Phi}}$ [e.g., at $t = 0.6$ ns for $L_0 = 34.9$ nm in **Figs. S11(c)** and **S11(e)**]. Moreover, the temporal evolutions of the growing dry patches (L) show that all simulated data on hydrophobic surfaces with different lengths ($L_0 = 16.8 - 34.9$ nm) collapse onto one master curve [**Fig. S11(f)**], following a power law with an exponent $2/3$, regardless of whether there is a wetting scenario.

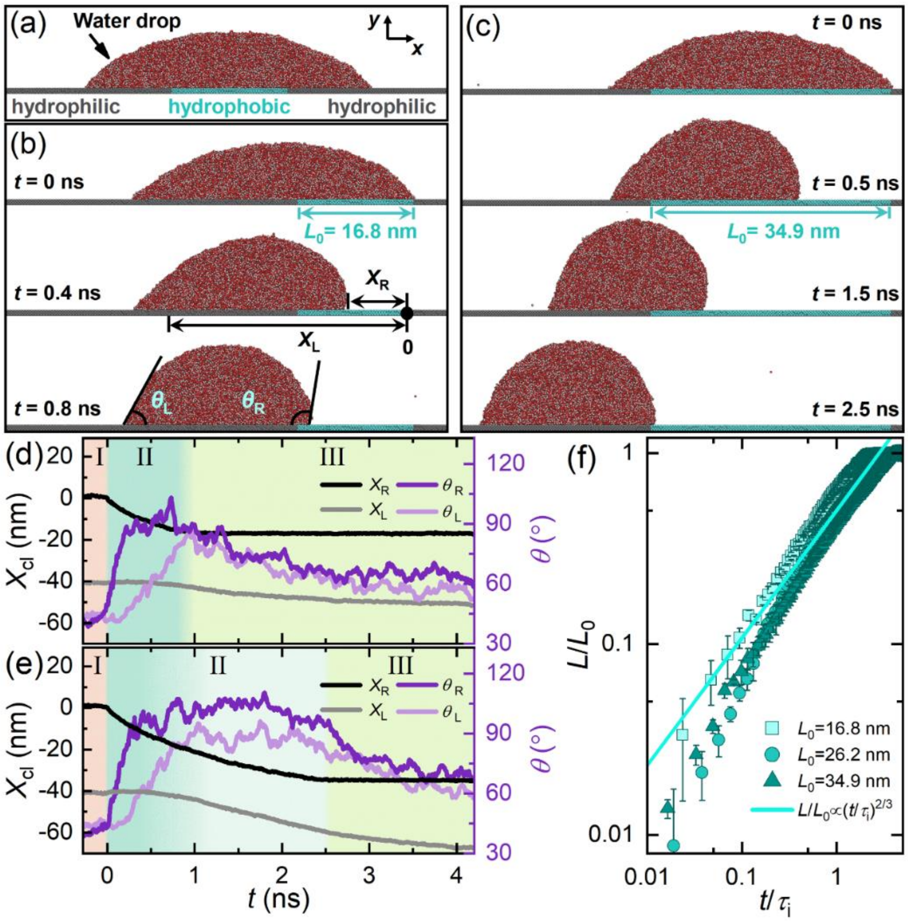

Fig. S11. (a) A pre-equilibrated water nanodrop consisting of $36{,}015$ water molecules on a hydrophobic–hydrophilic strip with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 114^{\circ}$, $\theta_{\mathrm{eq}}^{\mathrm{Phi}} \approx 36^{\circ}$ and $L_0 = 16.8\ \mathrm{nm}$. Oxygen atoms: red, hydrogen atoms: white, solid atoms in the hydrophobic part: cyan, and in the hydrophilic part: gray. (b–c) Snapshots of a dewetting water nanodrop from the hydrophobic zone ($\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 113.9 \pm 0.2^{\circ}$) with $L_0 = 16.8\ \mathrm{nm}$ (b) and $L_0 = 34.9\ \mathrm{nm}$ (c). (d–e) Temporal evolution of the instantaneous left (X_{L}) and right (X_{R}) contact line positions and contact angles on the left (θ_{L}) and right (θ_{R}) at $L_0 = 16.8\ \mathrm{nm}$ (d) and $L_0 = 34.9\ \mathrm{nm}$ (e). (f) Normalized length of the dry patch L/L_0 versus nondimensionalized time t/τ_{i} on hydrophobic strips with three different lengths. The solid line shows $L/L_0 \sim (t/\tau_{\mathrm{i}})^{2/3}$.

Supplementary Section VIII. Showcasing the stochastic dewetting from either side of contact line

Since no external forces were applied on the established dewetting systems during the entire molecular dynamics simulations, the dynamic dewetting is thus initially driven and triggered by molecular thermal motions. Due to the stochasticity of the molecular thermal motions, the dewetting can start on either side of the quasi-two-dimensional nanodrop, which have indeed observed in our simulations. In **Figs. 3(b)–3(c)** of the main text, we show the cases of moving nanodrop from right to left. Here, we provide snapshots of a dewetting process from left to right in **Fig. S12**, and similar dewetting processes as described in the main text [**Figs. 3(b)–3(c)**] have been observed.

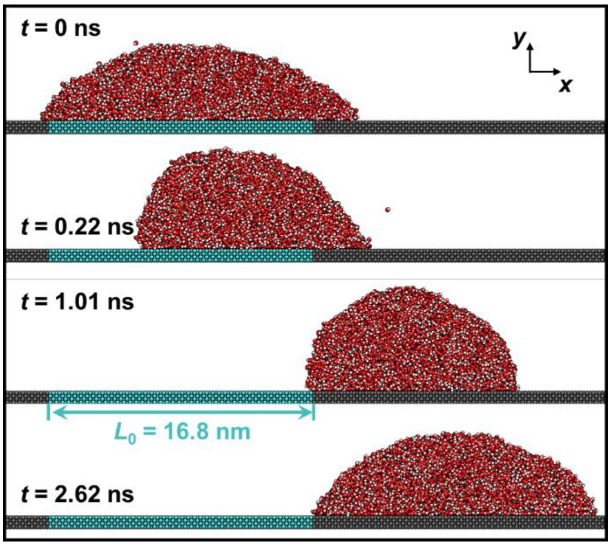

Fig. S12. Selected snapshots showing nanodrop dewetting from the left side to right side.

Supplementary Section IX. Identification of water molecules near the moving contact line

To identify the motion of water molecules near the moving contact line and to color these water molecules for clarifying the mode of contact line motion [**Fig. S13**(**a**)], we need first to determine the characteristic size of the solid–liquid interface in the simulations. By computing the density profile along the direction normal to the solid surface, as shown in **Fig. S13**(**b**), we can find the extensively-reported liquid density fluctuations when liquids meet solids. Though the amplitudes of density fluctuations depend on the surface wettability (i.e, θ_{eq} or ε), water density (ρ) was always found to approach to the bulk value at $y \gtrsim 1.0\ \mathrm{nm}$ on "hydrophobic" zone with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} = 113.9 \pm 0.2°$ and hydrophilic surface with $\theta_{\mathrm{eq}}^{\mathrm{Phi}} = 47.4 \pm 0.7°$. Therefore, we defined the water region within $1.0\ \mathrm{nm}$ above the solid surface as the solid–liquid interface [colored yellow in **Fig. S13(a)**]. Similarly, we computed the lateral density distribution of water molecules within the solid–liquid interface [**Fig. S13(c)**], and a remarkable density reduction was observed at the three-phase contact line. By fitting the density profile along the lateral dimension with a sigmoid function $\rho = k\left[1 + \tanh\left(\frac{x_{\mathrm{CL}} - x}{w_{\mathrm{CL}}/2}\right)\right]$, we obtained the characteristic width of the contact line w_{CL}, which is $6.2 \pm 0.3\ \text{Å}$ and $8.4 \pm 0.3\ \text{Å}$ on "hydrophobic" zone with $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 114°$ and hydrophilic surface with $\theta_{\mathrm{eq}}^{\mathrm{Phi}} \approx 47°$, respectively. By locating the right/left most water molecules in the solid–liquid interface (i.e., x_{R} and x_{L} in the main text) and drawing a sector of radius w_{CL} and of center x_{R} or x_{L}, as shown in **Fig. S13**(**a**), we identified these water molecules within this sector as the three-phase confluence regime [6, 7] [colored green in **Fig. 4(a)** and **Movie 3**]. And outermost water molecules of a shell thickness $1.0\ \mathrm{nm}$ were defined as the liquid–gas interface [colored orange in **Fig. 4(a)**, **Fig. S13(a)** and **Movie 3**].

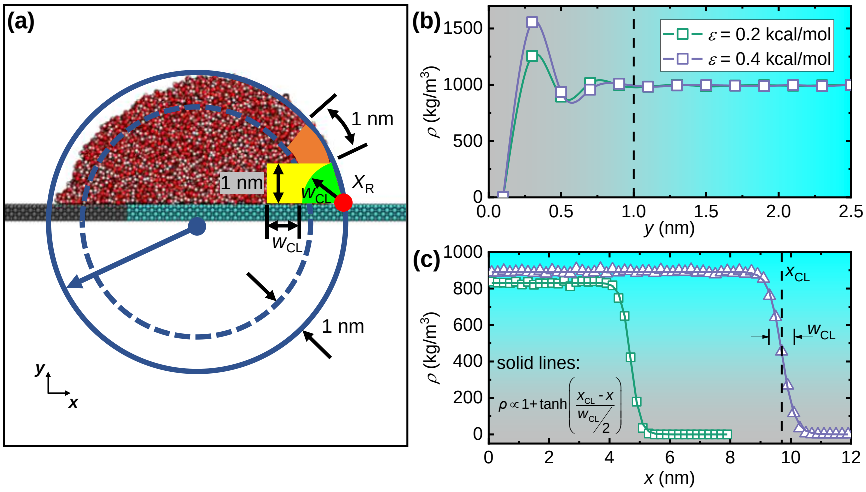

Fig. S13. Defining interfaces by water molecule density. (a) Schematic of identifying water molecules in the contact line (green), in the solid–liquid interface (yellow) and in the liquid–gas interface (orange). (b) Spatial distribution of water molecule density along the direction normal to the solid surface. (c) Spatial distribution of water molecule density along the lateral direction.

Supplementary Section X. Capillary-inertial wetting of water droplets on solid surfaces

Liquid wetting experiments were conducted by depositing pure water droplets with radii $R_0 \approx 0.9\ \mathrm{mm}$ on the as-prepared solid surfaces with $\theta_{\mathrm{eq}} \approx 80°$ and $102°$. Droplet spreading processes were recorded by a high-speed camera at $20{,}000\ \mathrm{fps}$ from the side view. As shown in **Figs. S14(a)** and **S14(b)**, upon touching the solid surface, water droplets immediately spread out to maximize the liquid–solid contact. In **Fig. S14(c)**, the nondimensional wetting radius R_{W}/R_0 is plotted as a function of the time normalized by the characteristic capillary-inertial time t/τ_{i} in the log-log coordinate, where $\tau_{\mathrm{i}} = \sqrt{\rho R_0^3/\gamma}$ is the characteristic capillary-inertial time. The linear dependence suggests that droplet spreading follows a power law $R_{\mathrm{W}}/R_0 \propto (t/\tau_{\mathrm{i}})^{\beta}$, and the linear regression analyses of these data provide $\beta \approx 0.41$ and 0.37 for spreading on solid surfaces with $\theta_{\mathrm{eq}} \approx 80°$ and $102°$, respectively.

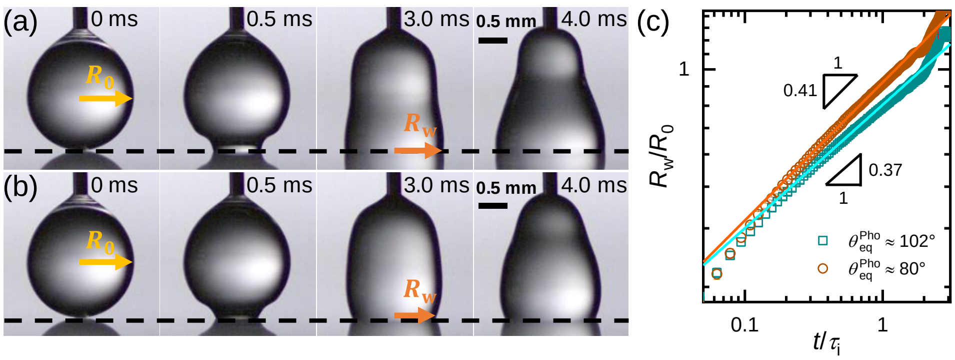

Fig. S14. Capillary-inertial wetting of water drops on solid surfaces. (a)–(b), Selected snapshots of a water droplet spreading on the solid surface with (a) $\theta_{\mathrm{eq}} \approx 80°$ and (b) $\theta_{\mathrm{eq}} \approx 102°$. (c) Plot of the nondimensional wetting radius R_{W}/R_0 as a function of the time normalized by the characteristic capillary-inertial time t/τ_{i} on solid surface. The solid lines are least-square fits with $R_{\mathrm{W}}/R_0 \propto (t/\tau_{\mathrm{i}})^{\beta}$, $\beta = 0.41$ and 0.37 for $\theta_{\mathrm{eq}} \approx 80°$ and $102°$, respectively.

Supplementary Tables

Table S1. Equilibrium (θ_{eq}), advancing (θ_{a}) and receding (θ_{r}) contact angles of 4 μL water droplets on diverse solid surfaces, and the Cassie–Baxter theoretical (θ_{the}) contact angles. θ_{the} was calculated by the well-known Cassie-Baxter relation for composite surfaces: $\cos\theta_{the} = f_{Phi}\cos\theta_{eq}^{Phi} + f_{Pho}\cos\theta_{eq}^{Pho}$, where θ_{eq}^{Phi} and θ_{eq}^{Pho} are the equilibrium contact angles while f_{Phi} and f_{Pho} are the fractional surface areas of the hydrophilic and hydrophobic materials, respectively. It can be seen that both θ_{eq} and θ_{the} nearly change with φ , suggesting that patterning hydrophilic surfaces with hydrophobic zones does not notably change the wetting properties of the original surfaces, because of the small φ.

Substrates	Name of surfaces	R_0 (μm)	S (μm)	φ	θ_{eq} (°)	θ_{a} (°)	θ_{r} (°)	θ_{the} (°)
Pure glass	Glass	NA	NA	0	42 ± 2	65 ± 3	16 ± 3	NA
Hydrophobized glass surface by silanizing for 10 h	Glass-10h	NA	NA	1	102 ± 1	113 ± 2	87 ± 5	NA
Hydrophobized glass by silanizing for 85 min	Glass-85 min	NA	NA	1	80 ± 1	96 ± 2	65 ± 1	NA
Hybrid hydrophilic–hydrophobic glass surfaces (selectively hydrophobized by silanizing for 10 h)	Glass-R10-10h	10	160	0.01	43 ± 1	62 ± 2	18 ± 2	43
	Glass-R25-10h	25	160	0.05	43 ± 1	65 ± 2	15 ± 1	46
	Glass-R50-10h	50	400	0.03	44 ± 2	67 ± 2	16 ± 1	45
	Glass-R100-10h	100	1600	0.01	45 ± 2	66 ± 2	17 ± 2	43
	Glass-R150-10h	150	700	0.08	45 ± 2	68 ± 5	18 ± 3	48
	Glass-R200-10h	200	3200	0.01	44 ± 1	66 ± 3	16 ± 3	43
Hybrid hydrophilic–hydrophobic glass surfaces (selectively hydrophobized by silanizing for 85 min)	Glass-R10-85min	10	160	0.01	42 ± 1	62 ± 1	15 ± 2	43
	Glass-R25-85min	25	160	0.05	45 ± 1	66 ± 5	15 ± 1	44
	Glass-R50-85min	50	400	0.03	43 ± 1	62 ± 1	17 ± 1	44
	Glass-R100-85min	100	1600	0.01	42 ± 2	60 ± 3	15 ± 2	43
	Glass-R150-85min	150	700	0.08	43 ± 2	65 ± 3	16 ± 1	45
	Glass-R200-85min	200	3200	0.01	42 ± 1	59 ± 2	13 ± 1	43

Table S2. Existing power-laws $R_\mathrm{w}^* \sim t^{*\beta}$ for capillary-inertial wetting of water droplets on solid surfaces, where $R_\mathrm{w}^* = R_\mathrm{W}/R_0$ is the nondimensional spreading radius, and t^* is the nondimensional spreading time.

Driving force	Resisting force	β	Surface wettability (θ_eq)	Timescale	References
Capillary force	Inertia force	$\frac{1}{2}$	0°	10 ms	Biance *et al.* [8]
Capillary force	Inertia force	$\frac{1}{4}-\frac{1}{2}$	2–115°	5.7 ms	Bird *et al.* [9]
Capillary force	Inertia force	$\frac{3}{10}-\frac{1}{2}$	0–112°	10 ms	Chen *et al.* [10]
Capillary force	Inertia force	0.41	80°	8 ms	This work (Supplementary Section 10)
Capillary force	Inertia force	0.37	102°	8 ms	This work (Supplementary Section 10)

Table S3. The characteristic capillary-inertial time τ_i, the characteristic capillary-inertial velocity V_i, the characteristic capillary-viscous time τ_v and the characteristic capillary-viscous velocity V_v for water dewetting on diverse "hydrophobic" zones with different radii R_0 and wetting properties $\theta_\mathrm{eq}^\mathrm{Pho}$.

Wettability of the hydrophobic zone	R_0 (μm)	τ_i (ms)	V_i (m/s)	τ_v (ms)	V_v (m/s)
$\theta_\mathrm{eq}^\mathrm{Pho} \approx 102°$	10	0.03	0.68	0.0005	41.52
	25	0.07	0.68	0.0030	16.61
	50	0.15	0.68	0.0120	8.31
	100	0.29	0.68	0.0482	4.15
	150	0.44	0.68	0.1084	2.77
	200	0.58	0.68	0.1927	2.08
$\theta_\mathrm{eq}^\mathrm{Pho} \approx 80°$	10	0.02	0.80	0.0007	27.87
	25	0.06	0.80	0.0045	11.15
	50	0.12	0.80	0.0179	5.57
	100	0.25	0.80	0.0718	2.79
	150	0.37	0.80	0.1615	1.86
	200	0.50	0.80	0.2871	1.39

Supplementary Movies

Supplementary Movie 1 A representative dewetting process on a hydrophilic surface ($\theta_{\mathrm{eq}}^{\mathrm{Phi}} \approx 42°$) with a circular hydrophobic zone of $R_0 = 150\ \mu\mathrm{m}$, $S = 700\ \mu\mathrm{m}$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 102°$. The movie was recorded at $100\ \mathrm{fps}$ for the slow dewetting regime and at $100{,}000\ \mathrm{fps}$ for the fast dewetting regime.

Supplementary Movie 2 Representative dynamics of water nanodrops on two hybrid hydrophilic-hydrophobic surfaces ($\theta_{\mathrm{eq}}^{\mathrm{Phi}} \approx 47°$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 114°$) with different lengths of hydrophobic strips: $L_0 = 9.8\ \mathrm{nm}$ and $L_0 = 16.8\ \mathrm{nm}$. Nanodrop dynamics contain three regimes: the quasi-stationary regime I, the dewetting regime II and the relaxation regime III.

Supplementary Movie 3 Typical molecular motions during the dewetting process (i.e., Regime II of the drop dynamics) of a water nanodrop from a hybrid hydrophilic-hydrophobic surface ($\theta_{\mathrm{eq}}^{\mathrm{Phi}} \approx 47°$ and $\theta_{\mathrm{eq}}^{\mathrm{Pho}} \approx 114°$) with a hydrophobic strip of $L_0 = 16.8\ \mathrm{nm}$.

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